

Security Model and Access



Control Document

End-to-End Salesforce CRM Transformation for

Customer Operations and Service Management

5.1 Role Hierarchy Design

The GreenGrid Utilities role hierarchy is built directly around the High-Power and High-Interest stakeholders identified during the Milestone 1 phase.

The main goal of this architecture is to dynamically handle vertical data visibility. When Organization-Wide Defaults (OWD) are set to restrictive levels, the role hierarchy automatically opens up record visibility upward through the management line. This allows executive leadership to see real-time data and metrics without needing manual sharing rules.

To maintain consistency across all project assets, the Salesforce roles map exactly to the team definitions established in the project charter and process swimlanes:

- **Executive Management (Top Tier):** Positioned at the very top of the hierarchy. This role has complete, unrestricted access to all underlying Cases, asset operations, and technician metrics to feed their live management dashboards.
- **Customer Support Team (Baseline Tier):** Staff in this role manage incoming calls and create new records. They operate at a foundational level and are laterally blocked from viewing field technician records to keep their workspace focused.
- **Field Technicians (Baseline Tier):** Operating as a peer level to the support desk, this role is for remote, on-site workers. They own individual records directly on their mobile devices, and their data remains isolated from the support team's view.

5.2 Permission Sets Definition

Following the principle of **Least Privilege Access**, base user profiles are restricted to basic operational views. Advanced data management features are granted dynamically using Permission Sets. Since the system uses flexible Lookup Relationships instead of rigid

Master-Detail restrictions, Permission Sets are used to give field technicians independent record ownership and editing capabilities.

The system uses two dedicated permission set extensions:

5.2.1 Field Technician Mobile Access (Field_Technician_Mobile_Access)

This is assigned only to users in the Field Technicians role to allow real-time mobile validation and data entry directly from service locations.

Target Object	Object-Level Permission	Field-Level Access Parameter
Account	Read-Only	Limited to customer identity and billing location strings.
Case	Read-Only	Limited to historical descriptions and priority fields for context.
Utility_Asset__c	Read-Only	Read-Only view of equipment serial numbers and indexing keys.
Field_Visit__c	Read, Create, Update	Full editing rights for Repair_Status__c and Technician Notes fields.

5.2.2 Customer Support Escalation Permissions (Support_Desk_Escalation_Access)

This container is given to senior coordinators on the Customer Support Team who need elevated administrative control during critical grid outages.

Target Object	Object-Level Permission	Field-Level Access Parameter
Case	Read, Edit, Delete	Allows changing record ownership and updating escalation priority metrics.
Field_Visit__c	Read, Edit	Allows balancing technical work distribution across queues.

5.3 Sharing Rules Configuration

To prevent duplicate data and protect underlying utility parameters, standard baseline sharing settings are locked down. Automated sharing rules are used purely as exceptions to open up data lines based on business needs.

5.3.1 Organization-Wide Defaults (OWD) Baseline

Data security settings for standard and custom objects are configured as follows to safeguard information:

Object Name	API Name	OWD Setting	Protection Target
Account	Account	Private	Stops cross-regional customer browsing and duplication.
Contact	Contact	Private	Protects primary phone numbers and communication histories.
Case	Case	Private	Restricts active phone logs to the assigned support agent's view.
Utility Asset	Utility_Asset__c	Private	Protects master equipment serial numbers and grid asset nodes.
Field Visit	Field_Visit__c	Private	Restricts technical task details to assigned mobile operator queues.

5.3.2 Criteria-Based Automated Sharing Rule

To meet the business goal of reducing response times during emergencies, an automated sharing exception rule has been set up in the platform.

- **Rule Developer Name:** High_Priority_Case_Sharing
- **Object Type:** Case
- **Rule Type:** Based on criteria

- **Field Criteria Parameter:** Priority EQUALS High AND Status EQUALS New OR Working
- **System Action Sharing Target:** Automatically shares matching Case records with the **Executive Management** public role group with **Read/Write** permissions.
- **Business Justification:** This exception allows major grid infrastructure issues to bypass base data restrictions, giving executive leaders immediate visibility into critical problems.

5.4 Salesforce Implementation and Configuration

This section provides the step-by-step instructions for manually configuring this data control model in a Salesforce Developer Edition environment.

Step 1: Organization-Wide Defaults (OWD) Lockdown

Secure the system data access by changing the sharing configuration inside the Sharing Settings utility.

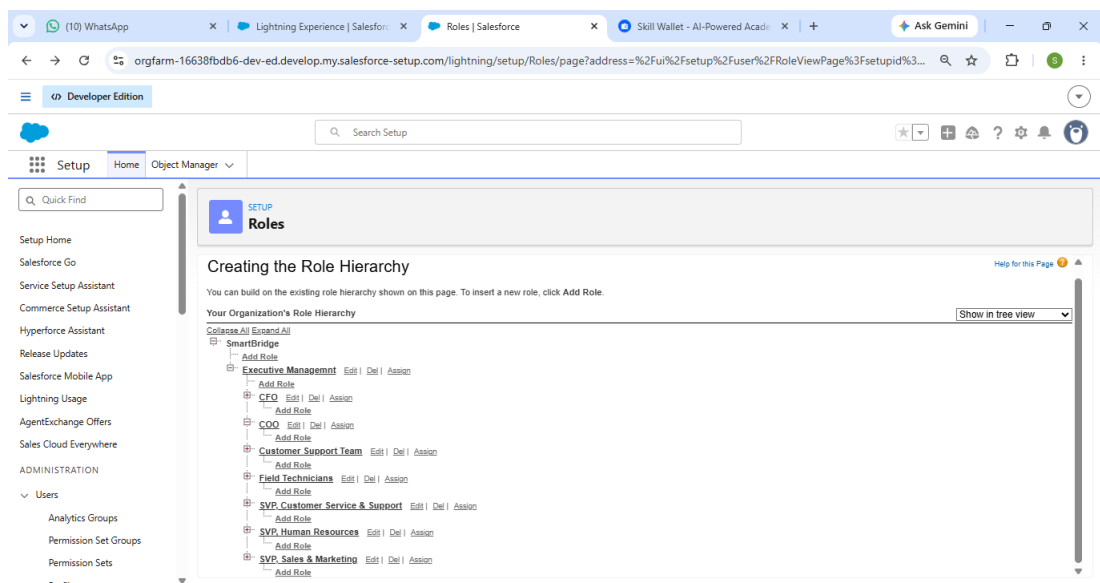
1. Go to **Setup, Security, Sharing Settings**.
2. Click **Edit** on the Organization-Wide Defaults section.
3. Change the **Default Internal Access** column to **Private** for the following objects: Account, Contact, Case, Utility_Asset__c, and Field_Visit__c.
4. Click **Save & OK**, then wait for the system confirmation email regarding data recalculation.

Object	Default Internal Access	Default External Access	Grant Access Using Hierarchies
Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Public Read/Write	Private	✓
Contact	Controlled by Parent	Controlled by Parent	✓
Order	Controlled by Parent	Controlled by Parent	✓
Asset	Controlled by Parent	Controlled by Parent	✓
Opportunity	Public Read/Write	Private	✓
Case	Public Read/Write/Transfer	Private	✓
Campaign	Public Full Access	Private	✓
Campaign Member	Controlled by Campaign	Controlled by Campaign	✓
User	Public Read Only	Private	✓
Activity	Private	Private	✓
Calendar	Hide Details and Add Events	Hide Details and Add Events	✓
Price Book	Use	Use	✓
Product	Public Read/Write	Public Read/Write	✓
Individual	Public Read/Write	Private	✓

Step 2: Role Hierarchy Structure Configuration

Set up the corporate organizational tree to control data visibility and automatic management roll-ups.

1. Go to **Setup, Users, Roles**.
2. Choose the corporate hierarchy view and click **Set Up Roles**.
3. Find the standard, default **CEO** role located right below the main company root node, and click **Edit**.
4. Rename the **Role Label** to Executive Management (the Role Name will automatically become Executive_Management). Click **Save**.
5. Click **Add Role** directly underneath the new Executive Management node to create the first sub-branch named Customer Support Team. Click **Save**.
6. Go back to the Executive Management node, click **Add Role** again to create the parallel sub-branch named Field Technicians, and click **Save**.



Step 3: Custom Permission Set Construction

Create a specialized permission container to give field operations extra object and field-level access without changing their base profiles.

1. Go to **Setup, Users, Permission Sets** and click **New**.
2. Label the permission set Field Technician Mobile Access (API Name: Field_Technician_Mobile_Access) and click **Save**.

The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar has a search bar with 'permi' and a list of navigation items: Users, Permission Set Groups, Permission Sets (selected), Custom Code, and Custom Permissions. The main content area is titled 'Permission Sets' and 'Create'. It contains a form with the following sections:

- Enter permission set information:**
 - Label: Field Technician Mobile Access
 - API Name: Field_Technician_Mobile_Access
 - Description: (empty text area)
 - Session Activation Required: ☐
- Select the type of users who will use this permission set:**
 - Who will use this permission set? (dropdown menu)
 - Instructions:
 - Choose --None-- if you plan to assign this permission set to multiple users with different user and permission set licenses.
 - Choose a specific user license if you want users with only one license type to use this permission set.
 - Choose a specific permission set license if you want this permission set license auto-assigned with the permission set.
 - Not sure what a permission set license is? [Learn more here.](#)
 - License: (dropdown menu)

3. Open **Object Settings**, scroll down, select **Field Visits**, and click **Edit**.
4. Check the boxes for **Read**, **Create**, and **Edit**. Under Field Permissions, check **Edit**

The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar has a search bar with 'permi' and a list of navigation items: Users, Permission Set Groups, Permission Sets (selected), Custom Code, and Custom Permissions. The main content area is titled 'Permission Sets' and 'Edit'. It contains a form with the following sections:

- Object Permissions:**

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☐
View All Records	☐
Modify All Records	☐
View All Fields	☐
- Field Permissions:**

Field Name	Field API Name	☐ Read Access	☐ Edit Access
Created By	CreatedById	☐	☐
Last Modified By	LastModifiedById	☐	☐
Owner	OwnerId	☐	☐
Parent Case	Parent_Case__c	☐	☐
Repair Status	Repair_Status__c	☒	☒
Resolution Timestamp	Resolution_Timestamp__c	☐	☐
Technician Notes	Technician_Notes__c	☒	☒
Visit Name	Name	☐	☐

5. **Access** for both Repair Status and Technician Notes.
6. Click **Save**.

Step 4: Automated Case Sharing Rule Deployment

Activate the automated data sharing exception to push high-priority cases to management queues.

1. Scroll down the main **Sharing Settings** page to the **Case Sharing Rules** related list and click **New**.
2. Label the rule High Priority Case Sharing. Set the **Rule Type** to **Based on criteria**.

3. Set the Criteria to: Priority EQUALS High.
4. Set **Share With** to: **Role, Executive Management**.
5. Set **Case Access** level to: **Read/Write**.

The screenshot shows the Salesforce Setup interface for configuring a Case Sharing Rule. The left sidebar shows the navigation menu with 'Setup' selected. The main content area is titled 'Sharing Settings' and 'Case Sharing Rule'. It includes a search bar and a 'Help for this Page' link. The configuration steps are as follows:

- Step 1: Rule Name**: Label is 'High Priority Case Sharing', Rule Name is 'High_Priority_Case_Sharing', and Description is empty.
- Step 2: Select your rule type**: Rule Type is 'Based on criteria' (selected).
- Step 3: Select which records to be shared**: Criteria table shows 'Priority' equals 'High'. Additional criteria are added with 'AND' logic.
- Additional Options**: 'Include records owned by users who can't have an assigned role' is checked.
- Step 4: Select the users to share with**: Share with is 'Roles' and 'Executive Management'.
- Step 5: Select the level of access for the users**: Case Access is 'Read/Write'.

Buttons for 'Save' and 'Cancel' are at the bottom.

6. Click **Save** (and click **OK** if a confirmation pop-up appears).

Step 5: Operational User Provisioning and Role Assignment

Create standard test users to verify that the Private OWD and vertical Role Hierarchy work correctly under live conditions.

1. Go to **Setup, Users, Users** and click the **New User** button.
2. Create the test user for the support desk environment:
 - **First Name:** Support | **Last Name:** Agent
 - **Email / Username:** Use a distinct format (e.g., support_@greengrid.dev).
 - **Role:** Assign to the Customer Support Team node.
 - **User License:** Salesforce
 - **Profile:** Standard User

New User

Salesforce now sends emails only from verified domains. To avoid delivery failures, verify your email domains or enable your substitute email-sending domain. [Learn More](#)

User Edit [Cancel] [Save] [Save & New]

General Information

First Name	Support	Role	Customer Support Team
Last Name	Agent	User License	Salesforce
Alias	sagen	Profile	Standard User
Email	syakilaan61+cs@gmail.co	Active	☒
Username	support_@greengrid.dev	Marketing User	☐
Nickname	User178038113264858080	Offline User	☐
Title		Knowledge User	☐
Company		Flow User	☐
Department		Service Cloud User	☐
Division		Site.com Contributor User	☐
		Site.com Publisher User	☐
		WDC User	☐
		Data.com User Type	None
		Data.com Monthly Addition Limit	Default Limit (300)
		Accessibility Mode (Classic Only)	☐

3. Click **Save & New**.

4. Create the test user for the remote mobile engineering workforce:

- **First Name:** Field | **Last Name:** Technician
- **Email / Username:** Use a unique format (e.g., field_@greengrid.dev).
- **Role:** Assign to the Field Technicians node.
- **User License:** Salesforce
- **Profile:** Standard User

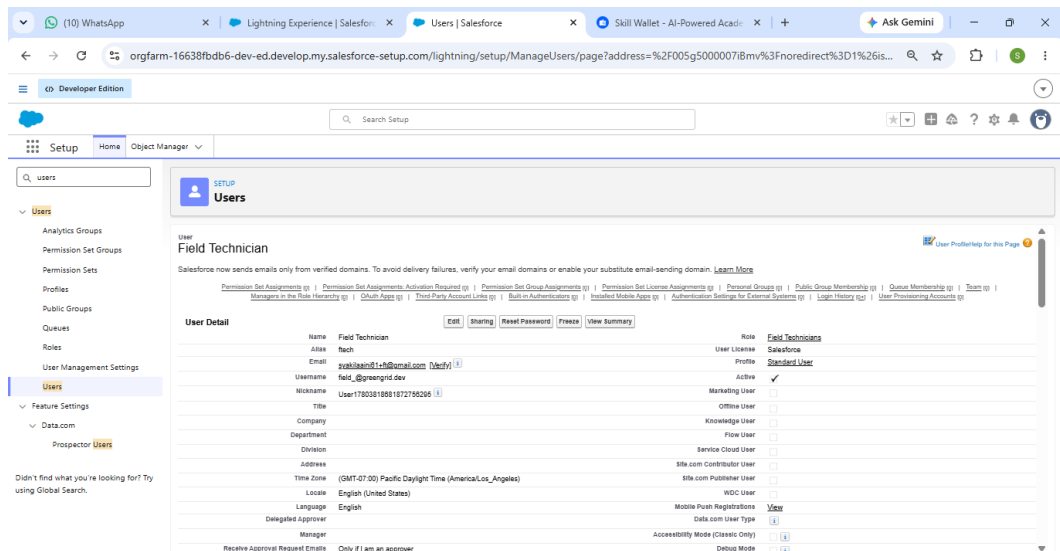
New User

Salesforce now sends emails only from verified domains. To avoid delivery failures, verify your email domains or enable your substitute email-sending domain. [Learn More](#)

User Edit [Cancel] [Save] [Save & New]

General Information

First Name	Field	Role	Field Technicians
Last Name	Technician	User License	Salesforce
Alias	ftech	Profile	Standard User
Email	syakilaan61+ft@gmail.com	Active	☒
Username	field_@greengrid.dev	Marketing User	☐
Nickname	User178038186818727562	Offline User	☐
Title		Knowledge User	☐
Company		Flow User	☐
Department		Service Cloud User	☐
Division		Site.com Contributor User	☐
		Site.com Publisher User	☐
		WDC User	☐
		Data.com User Type	None
		Data.com Monthly Addition Limit	Default Limit (300)
		Accessibility Mode (Classic Only)	☐



5. Click **Save**.

6. Go back to the **Active Users** list view to check that both the Support Agent and Field Technician entries are active, have the right profiles, and are assigned to the correct roles.

